

OAKLINE MARKET AWS INFRASTRUCTURE PROJECT 1

Oakline Market is a growing grocery business preparing to expand its online presence. The company requires a cloud infrastructure that supports its website, product image storage, staff access management, and future growth.

The objective of this project is to design, deploy, and document the AWS infrastructure required to support the business needs outlined below.

BUSINESS REQUIREMENTS

- A public-facing company website.
- Cloud storage for product images and marketing assets.
- Secure employee access management.
- Persistent storage for application data.
- High availability during periods of increased traffic.
- Scalability to support future business growth.

PHASE 1 – WEBSITE DEVELOPMENT

- Create a simple website for Oakline Market.
- Include: Company Name, About Us, Opening Hours, Contact Information, and at least one image.
- The homepage should clearly display: Welcome to Oakline Market.

PHASE 2 – SECURITY & ACCESS MANAGEMENT (IAM)

- Create user accounts representing different business departments.
- Suggested users: marketing.user, operations.user, and admin.user.
- Permissions should follow the principle of least privilege.

PHASE 3 – PRODUCT IMAGE STORAGE (S3)

- Create an S3 bucket for product images and marketing assets.
- Upload at least five product images.
- Document bucket settings and permissions.

PHASE 4 – WEBSITE DEPLOYMENT (EC2)

- Launch an EC2 instance and deploy the website created in Phase 1.
- Verify that the website is accessible through a browser.

PHASE 5 – PERSISTENT STORAGE (EBS)

- Create and attach an EBS volume to the EC2 instance.
- Document the volume size, attachment process, and purpose.

PHASE 6 – HIGH AVAILABILITY (ELASTIC LOAD BALANCER)

- Deploy two EC2 instances running the website.
- Configure an Elastic Load Balancer to distribute incoming traffic.
- Verify that traffic is routed through the load balancer.

PHASE 7 – SCALABILITY (AUTO SCALING GROUP)

- Create a Launch Template and an Auto Scaling Group.
- Suggested configuration: Minimum 2 instances, Maximum 4 instances.
- Ensure the environment can scale based on demand.

PROJECT DELIVERABLES

- Architecture Diagram
- Website Screenshot
- IAM Configuration Screenshots
- S3 Configuration Screenshots
- EC2 Configuration Screenshots

- EBS Configuration Screenshots
- Load Balancer Configuration Screenshots
- Auto Scaling Group Configuration Screenshots
- Short project report summarizing the deployment

REFLECTION QUESTIONS

1. Why is IAM important in a cloud environment?
2. Why is S3 suitable for storing product images?
3. What problem does a Load Balancer solve?
4. What happens if one EC2 instance becomes unavailable?
5. How does Auto Scaling improve reliability and scalability?